

ABG Interpretation Worksheet # 1

1. pH: 7.28 CO₂: 52 HCO₃: 24

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

2. pH: 7.31 CO₂: 60 HCO₃: 29

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

3. pH: 7.36 CO₂: 55 HCO₃: 30

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

4. pH: 7.50 CO₂: 30 HCO₃: 24

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

5. pH: 7.48 CO₂: 28 HCO₃: 20

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

6. pH: 7.44 CO₂: 30 HCO₃: 19

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

7. pH: 7.25 CO₂: 40 HCO₃: 18

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

8. pH: 7.29 CO₂: 32 HCO₃: 16

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

9. pH: 7.37 CO₂: 30 HCO₃: 18

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

10. pH: 7.52 CO₂: 40 HCO₃: 32

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

11. pH: 7.49 CO₂: 48 HCO₃: 31

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

12. pH: 7.43 CO₂: 48 HCO₃: 30

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

ABG Interpretation Worksheet # 2

13. pH: 7.40 CO₂: 40 HCO₃: 24

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

14. pH: 7.27 CO₂: 50 HCO₃: 23

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

15. pH: 7.51 CO₂: 29 HCO₃: 23

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

16. pH: 7.24 CO₂: 39 HCO₃: 17

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

17. pH: 7.53 CO₂: 41 HCO₃: 33

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

18. pH: 7.35 CO₂: 44 HCO₃: 24

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

19. pH: 7.30 CO₂: 58 HCO₃: 25

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

20. pH: 7.32 CO₂: 57 HCO₃: 28

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

21. pH: 7.38 CO₂: 50 HCO₃: 29

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

22. pH: 7.47 CO₂: 31 HCO₃: 22

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

23. pH: 7.46 CO₂: 29 HCO₃: 19

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

24. pH: 7.42 CO₂: 32 HCO₃: 20

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

ABG Interpretation Worksheet # 3

25. pH: 7.22 CO₂: 42 HCO₃: 16

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

26. pH: 7.30 CO₂: 33 HCO₃: 17

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

27. pH: 7.36 CO₂: 34 HCO₃: 19

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

28. pH: 7.50 CO₂: 43 HCO₃: 31

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

29. pH: 7.48 CO₂: 47 HCO₃: 30

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

30. pH: 7.41 CO₂: 47 HCO₃: 29

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

31. pH 7.28 / PaCO₂ 40 / HCO₃ 18

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

32. pH 7.33 / PaCO₂ 48 / HCO₃ 25

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

33. pH 7.37 / PaCO₂ 52 / HCO₃ 30

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

34. pH 7.48 / PaCO₂ 42 / HCO₃ 31

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

35. pH 7.44 / PaCO₂ 31 / HCO₃ 20

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

36. pH 7.40 / PaCO₂ 40 / HCO₃ 25

ABG Reading	Acid/Base/Normal?	
pH: Acid/Base of Blood		<----- 7.35 ----- 7.45 -----> ACID BASE
CO₂: Respiratory		<----- 35 ----- 45 -----> BASE ACID
HCO₃: Metabolic (Kidneys)		<----- 22 ----- 26 -----> ACID BASE
Interpretation:		

Answers with Rationale Workthrough

- 1 pH 7.28=Acid | CO2 52=Acid | HCO3 24=Normal → Respiratory Acidosis (Uncomp.)
- 2 pH 7.31=Acid | CO2 60=Acid | HCO3 29=Base → Respiratory Acidosis (Partial Comp.)
- 3 pH 7.36=Normal(acid lean) | CO2 55=Acid | HCO3 30=Base → Respiratory Acidosis (Full Comp.)
- 4 pH 7.50=Base | CO2 30=Base | HCO3 24=Normal → Respiratory Alkalosis (Uncomp.)
- 5 pH 7.48=Base | CO2 28=Base | HCO3 20=Acid → Respiratory Alkalosis (Partial Comp.)
- 6 pH 7.44=Normal(base lean) | CO2 30=Base | HCO3 19=Acid → Respiratory Alkalosis (Full Comp.)
- 7 pH 7.25=Acid | CO2 40=Normal | HCO3 18=Acid → Metabolic Acidosis (Uncomp.)
- 8 pH 7.29=Acid | CO2 32=Base | HCO3 16=Acid → Metabolic Acidosis (Partial Comp.)
- 9 pH 7.37=Normal(acid lean) | CO2 30=Base | HCO3 18=Acid → Metabolic Acidosis (Full Comp.)
- 10 pH 7.52=Base | CO2 40=Normal | HCO3 32=Base → Metabolic Alkalosis (Uncomp.)
- 11 pH 7.49=Base | CO2 48=Acid | HCO3 31=Base → Metabolic Alkalosis (Partial Comp.)
- 12 pH 7.43=Normal(base lean) | CO2 48=Acid | HCO3 30=Base → Metabolic Alkalosis (Full Comp.)
- 13 pH 7.40=Normal | CO2 40=Normal | HCO3 24=Normal → Normal ABG
- 14 pH 7.27=Acid | CO2 50=Acid | HCO3 23=Normal → Respiratory Acidosis (Uncomp.)
- 15 pH 7.51=Base | CO2 29=Base | HCO3 23=Normal → Respiratory Alkalosis (Uncomp.)
- 16 pH 7.24=Acid | CO2 39=Normal | HCO3 17=Acid → Metabolic Acidosis (Uncomp.)
- 17 pH 7.53=Base | CO2 41=Normal | HCO3 33=Base → Metabolic Alkalosis (Uncomp.)
- 18 pH 7.35=Normal | CO2 44=Normal | HCO3 24=Normal → Normal ABG
- 19 pH 7.30=Acid | CO2 58=Acid | HCO3 25=Normal → Respiratory Acidosis (Uncomp.)
- 20 pH 7.32=Acid | CO2 57=Acid | HCO3 28=Base → Respiratory Acidosis (Partial Comp.)
- 21 pH 7.38=Normal(acid lean) | CO2 50=Acid | HCO3 29=Base → Respiratory Acidosis (Full Comp.)
- 22 pH 7.47=Base | CO2 31=Base | HCO3 22=Normal → Respiratory Alkalosis (Uncomp.)
- 23 pH 7.46=Base | CO2 29=Base | HCO3 19=Acid → Respiratory Alkalosis (Partial Comp.)
- 24 pH 7.42=Normal(base lean) | CO2 32=Base | HCO3 20=Acid → Respiratory Alkalosis (Full Comp.)
- 25 pH 7.22=Acid | CO2 42=Normal | HCO3 16=Acid → Metabolic Acidosis (Uncomp.)
- 26 pH 7.30=Acid | CO2 33=Base | HCO3 17=Acid → Metabolic Acidosis (Partial Comp.)
- 27 pH 7.36=Normal(acid lean) | CO2 34=Base | HCO3 19=Acid → Metabolic Acidosis (Full Comp.)
- 28 pH 7.50=Base | CO2 43=Normal | HCO3 31=Base → Metabolic Alkalosis (Uncomp.)
- 29 pH 7.48=Base | CO2 47=Acid | HCO3 30=Base → Metabolic Alkalosis (Partial Comp.)
- 30 pH 7.41=Normal(base lean) | CO2 47=Acid | HCO3 29=Base → Metabolic Alkalosis (Full Comp.)
- 31 pH 7.28=Acid | CO2 40=Normal | HCO3 18=Acid → Metabolic Acidosis (Uncomp.)
- 32 pH 7.33=Acid | CO2 48=Acid | HCO3 25=Normal → Respiratory Acidosis (Uncomp.)
- 33 pH 7.37=Normal(acid lean) | CO2 52=Acid | HCO3 30=Base → Respiratory Acidosis (Full Comp.)
- 34 pH 7.48=Base | CO2 42=Normal | HCO3 31=Base → Metabolic Alkalosis (Uncomp.)
- 35 pH 7.44=Normal(base lean) | CO2 31=Base | HCO3 20=Acid → Respiratory Alkalosis (Full Comp.)
- 36 pH 7.40=Normal | CO2 40=Normal | HCO3 25=Normal → Normal ABG